

Solution Requirements

Date	02 July 2026
Team ID	SL-2026-02
Project Name	Smart Lender – Loan Eligibility Prediction Using Machine Learning
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Users can securely register and log in using a username and password before accessing the Smart Lender system.	High
2	Authorization Levels	The system provides role-based access for Loan Applicants and Administrators, ensuring appropriate permissions for each user.	High

3	External Interfaces	The application interacts with a trained Machine Learning model and applicant dataset stored in the system. It also supports data import/export in CSV format.	Medium
4	Transaction Processing	The system accepts applicant details, validates the input, processes the data through the ML model, and returns the loan eligibility prediction.	High
5	Reporting	The system displays loan prediction results and allows administrators to monitor prediction records and model performance.	Medium
6	Business Rules	Loan eligibility is determined based on applicant attributes such as income, credit history, education, employment status, loan amount, and property area using the trained ML model.	High
7	Compliance to Laws or Regulations	The system protects user information and follows standard data privacy and security practices for handling personal and financial data.	Medium
8	Other	The application provides a simple, responsive, and user-friendly interface with clear prediction results and error messages.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should generate loan eligibility predictions quickly after user submission.	Prediction completed within 2 seconds
2	Scalability	The application should support an increasing number of users and prediction requests without affecting performance.	Supports 500+ concurrent users
3	Security & Data Privacy	Applicant data should be securely stored and protected from unauthorized access using secure authentication and encrypted communication.	Secure Login and HTTPS communication
4	Reliability & Availability	The Smart Lender application should remain available with minimal downtime and provide consistent prediction results.	99.9% system availability
5	Usability & Accessibility	The application should provide an intuitive interface that is easy to use across desktops, laptops, tablets, and mobile devices.	Responsive UI with 95% user satisfaction

6	Other (Maintainability & Portability)	The application should be easy to maintain, update, and deploy on different operating systems and cloud platforms.	Modular code with cross-platform compatibility
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Summary:

The **Solution Requirements** define the functional and non-functional specifications of the **Smart Lender – Loan Eligibility Prediction Using Machine Learning** project. The functional requirements describe the core features, including secure authentication, role-based access, applicant data processing, loan eligibility prediction, reporting, and business rules. The non-functional requirements focus on system quality attributes such as performance, scalability, security, reliability, usability, and maintainability. Together, these requirements ensure that the Smart Lender system is accurate, secure, efficient, and capable of delivering a reliable loan prediction experience for users.